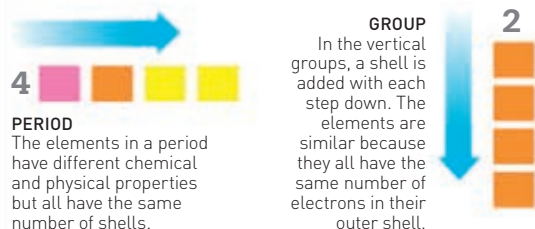


The table has horizontal rows called periods and vertical columns called groups. Atoms get bigger and heavier towards the bottom of each group because they have more protons and more electrons in the shells (rings) around them. As you move along the rows from left to right, atoms gain protons and electrons, and become more tightly packed.



**THE RUSSIAN SCIENTIST
DMITRI MENDELEEV
INVENTED THE
PERIODIC TABLE IN 1869**

- **Alkali metals**
Soft reactive metals, usually in compounds.
- **Alkaline earth metals**
Very reactive metals not found as pure elements.
- **Transition metals**
Malleable, ductile metals that are good conductors.

- **Rare earth metals**
Toxic, radioactive metals
often man-made.

Noble gases
Stable gases that do not react naturally.

Other metals
Malleable, ductile, solid,
dense metals.

Metalloids
Share properties with
metals and non-metals.

Non-metals
Poor solid conductors, brittle, with no metallic lustre, and gases.

Halogens
These exist in solid, liquid, and gas forms.

Unknown
Newly discovered
synthetic element.

READING THE TABLE

The table has horizontal rows called periods and vertical columns called groups. Atoms get bigger and heavier towards the bottom of each group because they have more protons and more electrons in the shells (rings) around them. As you move along the rows from left to right, atoms gain protons and electrons, and become more tightly packed.

4

PERIOD

The elements in a period have different chemical and physical properties but all have the same number of shells.

GROUP

In the vertical groups, a shell is added with each step down. The elements are similar because they all have the same number of electrons in their outer shell.

2

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PERIODIC TABLE KEY

Alkali metals

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Newly discovered synthetic element.

10

28

58.693

Ni



Spoon

NICKEL

11

29

63.546

Cu



Wire

COPPER

12

30

65.39

Zn



Sphalerite

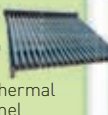
ZINC

13

31

69.723

Ga



Solar thermal panel

GALLIUM

14

32

72.64

Ge



Camera lens

GERMANIUM

15

33

74.922

As



Arsenic

ARSENIC

16

34

78.96

Se



Compact disc (CD)

TELLURIUM

17

35

79.904

Br



Bromine

BROMINE

18

36

83.80

Kr



Fluorescent lamp

KRYPTON

46

106.42

Pd



Catalytic converter

PALLADIUM

47

107.87

Ag



Silver pendant

SILVER

48

113

Cd



Yellow paint

CADMIUM

49

114.82

In



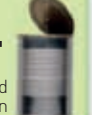
Light emitting diode (LED)

INDIUM

50

118.71

Sn



Tin-plated can

TIN

51

121.76

Sb



Bullet

ANTIMONY

52

127.60

Te



Compact disc (CD)

TELLURIUM

53

126.90

I



Liquid iodine

IODINE

54

131.29

Xe



Arc lamp

XENON

78

195.08

Pt



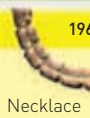
Ring

PLATINUM

79

196.97

Au



Necklace

GOLD

80

201

Hg



Thermometer

MERCURY

81

204.38

Tl



Rat poison

THALLIUM

82

207.2

Pb



Lead battery

LEAD

83

208.96

Bi



Crystal

BISMUTH

84

(209)

Po



Polonium

POLONIUM

85

(210)

At



Astatine

ASTATINE

86

(222)

Rn



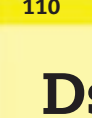
Radon

RADON

110

(281)

Ds



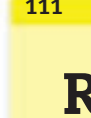
Darmstadtium

DARMSTADIUM

111

(272)

Rg



Roentgenium

ROENTGENIUM

112

285

Cn



Copernicium

COPERNICIUM

113

284

Uut



Ununtrium

UNUNTRIUM

114

289

Uuq



Ununquadium

UNUNQUADIUM

115

288

Uup



Ununpentium

UNUNPENTIUM

116

293

Uuh



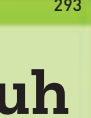
Ununhexium

UNUNHEXIUM

117

294

Uus



Ununseptium

UNUNSEPTIUM

118

294

Uuo



Ununoctium

UNUNOCTIUM

63

151.96

Eu



Xenotime

EUROPIUM

64

157.25

Gd



Gadolinium

GADOLINIUM

65

158.93

Tb



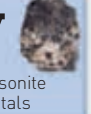
Green fluorescent lamp

TERBIUM

66

162.50

Dy



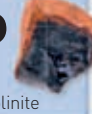
Fergusonite crystals

DYSPROSIUM

67

164.93

Ho



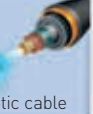
Gadolinite

HOLMIUM

68

167.26

Er



Fibre optic cable

ERBIUM

69

168.93

Tm



Thulium

THULIUM

70

173.04

Yb



Atomic clock

Ytterbium

YB

71

174.97

Lu



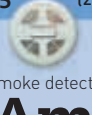
Lutetium

LUTETIUM

95

(243)

Am



Smoke detector

AMERICIUM

96

(247)

Cm



Curium

CURIUM

97

231.04

Bk



Berkelium

BERKELIUM

98

(251)

Cf



Californium

CALIFORNIUM

99

(252)

Es



Einsteinium

EINSTEINIUM

100

(257)

Fm



Fermium

FERMIUM

101

(258)

Md



Mendelevium

MENDELEVIUM

102

(259)

No



Nobelium

NOBELIUM

103

(262)

Lr



Lawrencium

LAWRENCIUM